The one-body factor of four: an audit and a lattice-matched continuum prescription

Research note for Gustav Delius

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Abstract

(c) The failed assertion in `audit_onebody.py` is an unjustified accuracy claim about finite-h extrapolations, not a failure of the limiting edge identity. Its maximum residual is $2.2546734882\,10^{-4}$, not below $3\,10^{-6}$. Independent inverse-Laplace calculations and fresh finite-sea solves diagnose the error.

(c) A more consequential omission is the continuum vacuum elastic renormalisation. In the specified good-equation prescription its sound-speed coefficient is $-9/(4\pi)$; under the *same formal continuation* as the soliton trace it is $+9/(4\pi)$ for the bad equation. Toda's harmonic vacuum gives $+3/(4\pi)$. Matching these independently calculated vacuum coefficients requires the finite Hamiltonian density $\Delta\mathcal{H} = -3\hbar_B u^2/(2\pi)$, and changes the continued soliton coefficient from $2/\pi$ to exactly $1/(2\pi)$. Both bad prescriptions then have the same sound-subtracted coefficient $-1/(4\pi)$.

(c) Conclusion **(A), for the leading coefficient in the stipulated continued trace**: a precise lattice-matched prescription exists, fixed by vacuum matching rather than a soliton fit. This is an effective-field-theory finite matching term, not just a different Wick contraction of u^3. **(d)** It does not construct the physical constrained bad-equation determinant, prove equivalence of complete quantum theories, or make the real stable-band trace agree with Toda. Those stronger identifications remain open.

1 Scope, conventions and checks

Labels are (a) proved in the literature, (b) conjectured or heuristic in the literature, (c) derived or checked here, and (d) unestablished identification or guess. Numerical (c) means evidence at the stated precision, not a rigorous error bound. No external article, abstract, DOI or arXiv entry is cited. The relevant internal sources are identified in Section 6. All scripts are in `phase2/scripts/`; each writes only its own `onebody_scheme_*.json` and ends with `ALL CHECKS PASSED`. We leave all existing files and git state alone.

(c) (`onebody_scheme_continuum.py`, `onebody_scheme_dictionary.py`) Use the stipulated potentials and Hamiltonians, with $u = \phi_x$, $\pi = \phi_t$, and $\Omega = \int \mathrm{d}\pi \wedge \mathrm{d}\phi$:

$$\mathcal{H}_g = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}u^2 - u^3 + \tfrac{1}{2}u_x^2, \qquad v_g = \sqrt{1-k^2}, \tag{1}$$

$$\mathcal{H}_b = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}u^2 + u^3 - \tfrac{1}{2}u_x^2, \qquad v_b = \sqrt{1+k^2}. \tag{2}$$

Here $k = \sin\alpha$ on the good branch and $k = \sinh\beta$ on the bad branch. We display right movers; parity supplies $|P|$ for left movers. For $u_0 = (k^2/2)\,\mathrm{sech}^2(kx/2)$ the checked integrals are

$$\int u_0 \mathrm{d}x = 2k, \qquad \int u_0^2 \mathrm{d}x = \frac{2}{3}k^3, \qquad \int u_0^3 \mathrm{d}x = \frac{4}{15}k^5, \qquad P_0 = v\int u_0^2 \mathrm{d}x. \tag{3}$$

Throughout F and W denote *coefficients* of $\hbar_B$ in the fixed-P soliton energy and homogeneous vacuum energy density, respectively. The notation s_1 means $c_{\mathrm{sound}} = 1 + \hbar_B s_1 + \cdots$ in fixed classical Boussinesq units. We test candidate schemes; none is adopted as the PI's preferred quantum theory.

2 Assessment of audit_onebody

2.1 The failed assertion: limiting identity versus extrapolation

(c) (`onebody_scheme_edge.py`) The original script was executed unchanged and failed at line 383, as reported. Its comparison is

$$C_v(y) = s_{1,T} + \frac{y}{2}\{u'(y) - 1\}, \qquad s_{1,T} = \frac{3}{4\pi}. \tag{4}$$

This is an $h \to 0$ *edge-limit* identity. The script substitutes independently extrapolated entries from `onebody_toda_results.json` and `frame_analysis.json`, with finite-h truncation errors. The input solves are double precision: evaluating the comparison at 65 digits does not improve their extrapolation accuracy. The central fits use only three h values for a quadratic intercept. At large y, the finite amplitude $\kappa^2 = hy/b$ is not uniformly small on those grids.

y	original signed residual	combined fit spread	finer quartic residual
8	$-7.5273\,10^{-7}$	$2.3259\,10^{-5}$	—
16	$-5.0951\,10^{-6}$	$8.4061\,10^{-5}$	$-1.6466\,10^{-9}$
32	$-3.4752\,10^{-5}$	$2.9805\,10^{-4}$	$-4.1939\,10^{-8}$
64	$-2.2547\,10^{-4}$	$1.1013\,10^{-3}$	$-9.3801\,10^{-7}$

(c) The residual is $C_v - s_1 - y(u'-1)/2$. Every original residual lies inside its recorded combined sensitivity spread. Fitting the finer saved finite-sea rows independently, with a quartic through the last seven h values down to .0025, recovers the original $3\,10^{-6}$ tolerance. At $y = 64$, a quintic gives $-2.2681\,10^{-7}$; a cubic gives $-5.5779\,10^{-6}$ and still fails that tolerance. These are model sensitivities, not certified bounds. We do not relax or edit the old check.

Independent calculation, not only refitting. **(c)** (`onebody_scheme_edge.py`) Factoring the half-line kernel at nonzero Fourier frequency gives the checked product and inverse transforms

$$g(s) = s^{-1/2}\Gamma\Big(1 + \frac{s}{2\pi}\Big)\exp\Big[-\frac{s}{2\pi}\log\frac{s}{2\pi e}\Big], \tag{5}$$

$$2\pi g(iw)g(-iw) = \frac{2\pi}{1 - e^{-|w|}}, \quad w \neq 0, \tag{6}$$

$$\mathcal{L}[F](s) = \sqrt{\pi}\, g(s)/s^2, \quad \mathcal{L}[f](s) = \sqrt{\pi}\, g(s)/s, \tag{7}$$

$$\mathcal{L}[J](s) = \sqrt{\pi}\, g(s)/s^3, \tag{8}$$

$$F(u(y)) = \frac{4}{3}y^{3/2}, \quad f = F', \quad J' = F. \tag{9}$$

The normalization is fixed by $F \sim 4u^{3/2}/3$, not by the target C_T. Fresh Talbot inversions and de Hoog contour checks at 45 digits agree within 10^{-35} in all three transforms. Differentiation and integration of the dressed momentum give

$$C_v = \frac{3}{4\pi} + \frac{3F}{4f} - \frac{y}{2}, \qquad C_E = \frac{3}{4\pi} + \frac{3J}{4F} - \frac{3y}{10}, \qquad u' = \frac{2\sqrt{y}}{f}. \tag{10}$$

Thus the fresh velocity results are

y	fresh C_v	old intercept minus fresh	finer intercept minus fresh
16	0.159293610890713	$-4.5672\,10^{-6}$	$-1.5142\,10^{-9}$
64	0.159189963972784	$-2.0347\,10^{-4}$	$-8.7287\,10^{-7}$

(c) Separate fresh composite-Gauss finite-sea solutions at $h = .02, .01, .005$ use $800, 1600, 3200$ nodes, with a 4000-node refinement at .005. They reproduce the finer saved velocities within 10^{-7}; the refinement changes the tested quantities by less than 10^{-11}. Their *finite-h* residual decreases towards zero. These computations use neither the old solver nor C_T as an input. This distinguishes truncation error from an incorrect limiting identity.

2.2 Six dictionary factors

(c) (`onebody_scheme_dictionary.py`) All six leading classical conversions have factor one. The following checks make clear what this assertion does and does not fix. Write a for mean Toda spacing and $b = e^{-a/2}$, with reference coordinates

$$x_B = 2\sqrt{3}\, n, \quad t_B = 2\sqrt{3}b\, t_T, \quad \phi_B = -U_T/\sqrt{3}, \quad S_T = 3bS_B, \quad \hbar_T = 3b\hbar_B. \tag{11}$$

Here U_T is displacement; u remains Boussinesq strain. The Taylor expansion of the Toda action through cubic strain and the first dispersive term, and the symplectic form, independently give the same factor $3b$.

1. **Planck constant.** The action factor fixes $\hbar_T = 3b\hbar_B$; there is no extra factor four in the semiclassical exponential.

2. **Momentum definitions.** The canonical soliton label momentum is $P_n = 4b(\kappa\cosh\kappa - \sinh\kappa)$, not the total particle momentum $\Pi_T = 2b\sinh\kappa$. The physical-frame canonical momentum is P_n/a. Their different small-κ orders prohibit interchanging them.

3. **Dressed momentum.** Independently integrate $p + (a\pi)^{-1}\int_0^\pi \Theta_0(p - 2b\cos t)\mathrm{d}t$, where $\Theta_0(d) = 2d(\log d - 1)$ and $p = 2b\cosh\kappa$. It equals P_n/a to better than 10^{-53} at four test points. At finite h, the checked origin is $P(q) = \pi h/a$; use $\overline{P} = P(p) - P(q)$.

4. **Energy and its origin.** Use $\overline{E} = E(p) - E(q)$ and $E_{T,0} = 2b^2(\sinh\kappa\cosh\kappa - \kappa)$. The conversion is
$$P_B = \frac{a\overline{P}}{6\sqrt{3}b}, \quad E_B = \frac{\overline{E}}{6\sqrt{3}b^2}, \quad \frac{\Delta E_B}{\hbar_B P_B} = \frac{3}{a}\frac{\Delta\overline{E}}{h\overline{P}}. \tag{12}$$
With $\kappa = \sqrt{3}\beta$, the leading momentum, energy, and fifth-order excess above the classical sound line all match. An energy-origin shift cannot change a velocity coefficient.

5. **Mass and width.** $M_B = 2k$, $M_T = -2\kappa$, and $M_T/(-\sqrt{3}M_B) \to 1$. Eliminating the width in favour of the canonical momentum removes a mere width reparametrisation.

6. **Scattering phase.** Independent evaluation of the two exact Hirota polynomials in overtaking and head-on channels gives $\delta_T/(3b\delta_b) \to 1$. At amplitude scale 10^{-5} the errors are $1.11\,10^{-10}$ and $6.49\,10^{-11}$. Division by the correspondingly scaled $\hbar$ leaves factor one.

(c) These checks exclude the proposed *multiplicative classical* explanations. They do not determine an additive order-$\hbar$ elastic counterterm, a quantum frame convention, or a projected determinant domain. The audit's conclusion that the dictionary rules out a scheme explanation does not follow.

2.3 Toda decomposition and qualifications to its proof

(c) (`onebody_scheme_dictionary.py`, `onebody_scheme_edge.py`) Fresh 60-digit finite-sea solves check the endpoint identity $D(q)V(q) = \pi h/a$, the edge momentum origin, and

$$v_s = -a\partial_a q\big|_h = \frac{a}{2}(q - h\partial_h q\big|_a), \qquad V(p) = \partial_a(aP(p))\big|_{p,h}. \tag{13}$$

At $h = .8, q = 1$, independent q and h differences with step 10^{-15} and 48 Gauss nodes satisfy the specified 10^{-27} tolerance. Endpoint products and derivatives are not evaluated by substituting these identities into themselves.

(c) (`onebody_scheme_continuum.py`) The Toda sound coefficient has an independent vacuum derivation. The harmonic frequency at spacing a is $\omega_T(K) = 2b|\sin(K/2)|$, so the ground-state energy per particle is

$$e(a,h) = b^2 + h\frac{1}{4\pi}\int_0^{2\pi}\omega_T(K)\mathrm{d}K + O(h^2) = b^2 + \frac{2hb}{\pi} + O(h^2). \tag{14}$$

With unit particle mass, the long-wave sound speed is $a\sqrt{\partial_a^2 e} = ab + ah/(4\pi) + O(h^2)$; the dictionary gives $s_{1,T} = 3/(4\pi)$ in Boussinesq units.

(c) (`onebody_scheme_edge.py`) Expanding (7) at small s and inverting the first logarithm gives

$$u(y) = y - \frac{\log y}{2\pi} + \frac{1 - \log(8\pi)}{2\pi} + O(y^{-1}\log^m y). \tag{15}$$

Insertion into (10) yields, for the stipulated nested edge limit,

$$C_T = \frac{3}{4\pi} - \frac{1}{4\pi} = \frac{1}{2\pi}. \tag{16}$$

Thus the decomposition is confirmed; "universal" here refers to this matched acoustic edge problem, not arbitrary UV completions or quantum frames. **(d)** The audit's claimed proof of commutation with a fixed nonzero-P semiclassical limit substitutes the large-y expansion into a different limit without a uniform remainder estimate. Its formal result is compatible with (16), but that substitution is not a proof of limit interchange.

Additional errors in the audit text. (c) (`onebody_scheme_dictionary.py`, `onebody_scheme_edge.py`) Theorem 3 prints $b = \pi/a$; the correct relation is $b = e^{-a/2}$. Its intermediate expression for $(V/D - v_s)/h$ incorrectly subtracts the classical $ay/6$ term: before subtracting classical soliton velocity the edge expression is $ayu'(y)/6$. The displayed final (4) is correct. If the inverse edge series contains e_0/y^2, integrating with $\mathrm{d}P \propto \sqrt{y}\mathrm{d}y$ gives the energy tail $3e_0/y^2$, not the printed $3e_0/(2y^2)$. None of these text errors changes the leading constant derived above.

2.4 What the short-wave claim establishes

(c) (`onebody_scheme_continuum.py`) The conditional good trace and its formal continuation are

$$F_g(k) = \frac{\sqrt{3}}{\pi}(\tfrac{3}{4} - k^2)\operatorname{artanh}(2k/\sqrt{3}) - \frac{3k}{2\pi}, \tag{17}$$

$$F_{\mathrm{ac}}(k) = -iF_g(ik) = \frac{\sqrt{3}}{\pi}(\tfrac{3}{4} + k^2)\arctan(2k/\sqrt{3}) - \frac{3k}{2\pi}. \tag{18}$$

They give $C_g = -2/\pi$ and $C_{\mathrm{ac}} = 2/\pi$. For the three algebraic roots $\varphi = \alpha, \alpha \pm 2\pi/3$, the coefficient of the logarithm in $e - vp$ is proportional to

$$w(\varphi) = 2\sin\varphi(\cos\alpha - \cos\varphi), \quad w(\alpha) = 0, \quad w(\alpha \pm 2\pi/3) = \pm 2\sqrt{3}(\tfrac{3}{4} - \sin^2\alpha). \tag{19}$$

This exactly confirms the *logarithmic* light-root cancellation. It does not prove that the spectral integral has support only near $|q| = \sqrt{3}/2$. In particular the elementary term of the light root in the displayed charge sum is $-k/\pi$, not zero; all elementary root terms together give $-3k/(2\pi)$. Allocation among roots, endpoint and tadpole is representation dependent. The accurate statement is that the final nonpolynomial term in (18) comes from the two heavy algebraic roots. The literal "100% short-wave modes" assertion overinterprets this algebra.

(c) (`onebody_scheme_continuum.py`) The audit also incorrectly describes the real-band resonance. At $k \to 0$, for a right mover and $q_* = -\sqrt{3}/2$,

$$\omega(q_*) = \frac{\sqrt{3}}{4}, \quad \omega'(q_*) = 1, \quad \omega(q_*) - q_* = \frac{3\sqrt{3}}{4} \neq 0. \tag{20}$$

It is a stationary co-moving frequency ($\omega' - v \to 0$), not a zero co-moving frequency $\omega = -vq$. The positive-q group derivative there is -1 and reverses sign at negative q; the audit's $-1/2$ is wrong. The maximum of $\omega(q)$ on the positive band is at $1/\sqrt{2}$, not $\sqrt{3}/2$. The latter lies on the descending part, still inside the stable band.

3 Vacuum calculation and a precise matching prescription

3.1 Good continuum: the missing acoustic renormalisation

(c) (`onebody_scheme_continuum.py`) Put a constant strain $u = U + \eta_x$ on the good vacuum. The quadratic frequency at fixed reference coordinates is

$$\omega_g(q;U) = |q|\sqrt{q^2 + 1 - 6U}. \tag{21}$$

Its zero-point energy coefficient at a fixed wave-number cutoff is

$$W^{\rm bare}_{g,\Lambda}(U) = \frac{1}{2\pi}\int_0^\Lambda q\sqrt{q^2+1-6U}\,\mathrm{d}q = \frac{(\Lambda^2+1-6U)^{3/2} - (1-6U)^{3/2}}{6\pi}. \tag{22}$$

Subtract the vacuum constant and exactly its first derivative at $U = 0$. The latter is the specified normal-ordering tadpole, since $(W^{\rm bare}_{g,\Lambda})'(0) = -3(\sqrt{1+\Lambda^2} - 1)/(2\pi)$. The remaining limit is

$$W_g(U) = -\frac{(1-6U)^{3/2} - 1 + 9U}{6\pi}, \qquad W_g''(0) = -\frac{9}{2\pi}. \tag{23}$$

The convergent curvature is independently checked by

$$W_g''(0) = -\frac{9}{2\pi}\int_0^\infty \frac{q\mathrm{d}q}{(1+q^2)^{3/2}} = -\frac{9}{2\pi}. \tag{24}$$

Two direct subtracted quadratures at finite $\Lambda = 100$ agree with the exact primitive within 10^{-62}. A linear tadpole subtraction has zero second derivative: it does not remove this finite bubble contribution.

(c) With canonical kinetic coefficient one, the low-frequency quadratic effective action therefore has elastic modulus $1 + \hbar_B W_g''(0)$ and

$$s_{1,g} = \tfrac{1}{2}W_g''(0) = -\frac{9}{4\pi}. \tag{25}$$

Each external potential fluctuation at a cubic vertex carries a spatial derivative. The static bubble fixes the elastic q^2 term, and the uniform kinetic mass density remains one. Equation (25) is the *thermodynamic acoustic coefficient* inferred from that elasticity; it is not an assertion that a perturbative on-shell phonon self-energy has a uniform expansion at an acoustic threshold. This directly refutes the audit's assertion $s_{1,\rm cont} = 0$ for the stated bare-normalized continuum calculation. Defining the *renormalized* speed to be one instead requires an additional finite subtraction, absent from that audit.

3.2 Continued bad vacuum and lattice vacuum

(c) /(d) (`onebody_scheme_continuum.py`) Continue the *same prescription* as $F_b(k) = -iF_g(ik)$, rather than treating unstable modes as a real vacuum. The local density rule is

$$W_{b,\rm ac}(U) = -W_g(-U) = \frac{(1+6U)^{3/2} - 1 - 9U}{6\pi}, \qquad s_{1,b,\rm ac} = \frac{9}{4\pi}. \tag{26}$$

The sign is fixed by the spatial continuation as well as $U_g = -U_b$: for the branch $k_g = ik_b$ one has $x_g = -ix_b$, and $-i\int \mathrm{d}x_g W_g(U_g) = -\int \mathrm{d}x_b W_g(-U_b)$. Equivalently, retain the real analytic finite part of the integrated $q\sqrt{1+6U-q^2}$ primitive, then subtract its constant and linear terms. The algebra is (c); interpreting this as a physical bad-equation vacuum is (d). A fixed real stable-band cutoff is a different prescription and does not justify this continuation.

(c) (`onebody_scheme_continuum.py`) The Toda counterpart is real and requires no continuation. Keep the reference b and coordinates of (11) fixed. A homogeneous Boussinesq strain

changes lattice spacing by $r - a = -6U$, so the harmonic frequency in (14) is multiplied by e^{3U}. The conversion of energy density is $1/(36b^2)$ and that of h is $3b$. Subtracting constant and linear terms gives

$$W_T^B(U) = \frac{e^{3U} - 1 - 3U}{6\pi}, \qquad \tfrac{1}{2}(W_T^B)''(0) = \frac{3}{4\pi}. \tag{27}$$

The linear subtraction fixes background pressure/topological energy; it does not affect the sound coefficient. This is the continuum local limit of the *lattice's own harmonic mode sum*, not a fitted soliton number.

3.3 Matching the elastic modulus

(c) (`onebody_scheme_continuum.py`) Equations (26) and (27) determine

$$W_T^B - W_{b,\mathrm{ac}} = \frac{e^{3U} - (1+6U)^{3/2} + 6U}{6\pi} = -\frac{3}{2\pi}U^2 + O(U^3). \tag{28}$$

Here is the precise prescription required only to this order in amplitude:

1. Keep the conditional equal-mode-count, exactly tadpole-subtracted trace and its continuation (18), with the stated momentum and energy origins.

2. Match the vacuum elastic modulus in the same reference coordinates to the microscopic Toda harmonic mode sum. Equivalently add

$$\boxed{\Delta\mathcal{H} = \hbar_B \lambda u^2, \qquad \lambda = s_{1,T} - s_{1,b,\mathrm{ac}} = -\frac{3}{2\pi}.} \tag{29}$$

   No order-$\hbar_B$ kinetic or canonical-momentum rescaling is introduced.

3. Evaluate this term on the classical soliton when computing the first-order energy at fixed canonical P. Profile changes do not contribute at this order, since the unperturbed profile is stationary for $H - vP$.

(c) (`onebody_scheme_continuum.py`) By (3) the resulting trace is

$$F_{\mathrm{matched}}(k) = F_{\mathrm{ac}}(k) - \frac{k^3}{\pi}, \qquad \boxed{\lim_{k\to 0} \frac{F_{\mathrm{matched}}(k)}{P_0(k)} = \frac{1}{2\pi}.} \tag{30}$$

The equality is exact for the *limiting coefficient*. We do not assert that this function at finite k equals Toda's dispersion. The full local difference in (28), as well as derivative matching terms, would require higher-amplitude matching; $\int u_0^3 = O(k^5)$ already shows why it cannot affect the present k^3 coefficient.

Natural or tuned? **(c)** The coefficient in (29) is forced if "lattice compatible" means matching the quantum vacuum elasticity of the specified Toda Hamiltonian, at fixed mean strain and in the inherited units. It was computed from the two vacuum mode sums, without using either soliton coefficient. This is natural finite matching of a low-energy parameter. It is not forced by the truncated Boussinesq equation alone. If one simply postulated this counterterm without its vacuum condition, it would be a tuned finite subtraction.

(c) The distinction is material. The lattice classical local potential in these units is $(e^{6U} - 1 - 6U)/36 = U^2/2 + U^3 + 3U^4/2 + \cdots$. Terms absent from the cubic continuum truncation and the full lattice dispersion participate in its quantum elasticity. Calling (29) a different *regularisation scheme* means matching these finite UV contributions into the effective Hamiltonian; it is not a claim that two fixed bare Hamiltonians have identical physical sound velocities. With renormalized parameters held fixed, a regulator change alone cannot change a physical prediction.

(c) The diagnostic cancellation is particularly transparent:

$$C_{\rm ac} - s_{1,b,\rm ac} = \frac{2}{\pi} - \frac{9}{4\pi} = -\frac{1}{4\pi} = C_T - s_{1,T}. \tag{31}$$

The audit subtracted Toda's sound shift but set the continuum one to zero. Its quoted ratio -8 therefore compares unequal acoustic baselines.

4 Other candidate prescriptions and the good equation

Cutoff endpoint and finite tadpole. **(c)** (`onebody_scheme_continuum.py`) In the good trace the required equal-mode-count endpoint has $B_\Lambda = 3k\Lambda/(2\pi) + O(\Lambda^{-1})$ and no finite constant term. Omitting it with the same tadpole leaves a divergence, as the earlier determinant audit found. A finite redefinition of that endpoint can encode an arbitrary finite subtraction, but no alternative endpoint quantization was derived here from the lattice. In particular merely shifting a cutoff is not an independent derivation of (29).

(c) A finite change to the Wick contraction of the cubic interaction adds a constant multiple of $M = 2k$. Since $P \sim 2k^3/3$, a nonzero such term creates a k^{-2} divergence in F/P. It cannot convert one finite coefficient into the other. Once the finite soft limit is imposed, that linear freedom is fixed. The u^2 matching term is different: it changes the elastic modulus and survives the old origin/mass annihilator $\partial_\beta^3 - \partial_\beta$ with limiting coefficient λ.

Real stable band (Scheme A). **(c)** (`onebody_scheme_continuum.py`) Fresh quadrature of the exact Hirota transmission, using $q = \sin t$ and an independently checked rational integrand in $z = \cos t$, confirms the stipulated continuous-half-band trace

$$F_{\rm band} = F_{\rm ac} - \frac{\sqrt{3}}{2}(k^2 + \tfrac{3}{4}), \qquad F_{\rm band}/P_0 \sim -\frac{9\sqrt{3}}{16k^3}. \tag{32}$$

The correct bad Wick term is $-3k/\pi$. Three 70-digit checks give errors below 10^{-60}. Subtracting the threshold constant alone still leaves an $O(k^2)$ obstruction. The elastic counterterm (29) does not remove either term. Consequently this specified Scheme A does *not* match Toda; projecting out further modes would require a new spectral domain and boundary prescription, not an unexplained deletion of (32)'s extra term.

What the lattice mode sum has and has not supplied. **(c)** It has supplied (27), and hence a non-tuned leading local matching condition. The independent Toda edge calculation supplies the remaining $-1/(4\pi)$, which agrees with the continued continuum trace after its own sound subtraction. **(d)** We have not constructed the lattice soliton fluctuation determinant and taken its complete continuum limit, or proved that its constrained measure and endpoints produce (30) at finite amplitude. Thus the present result is a solution for the requested leading coefficient, not a full regulator equivalence theorem.

Good equation. **(c)** (`onebody_scheme_continuum.py`) The well-posed good vacuum needs no unstable-band contour choice. Its finite elastic bubble is UV convergent after the stated linear subtraction; the prescribed value $s_{1,g} = -9/(4\pi)$ is not a UV divergence to discard. For its conditional soliton trace,

$$C_g = -\frac{2}{\pi}, \qquad C_g - s_{1,g} = +\frac{1}{4\pi}. \tag{33}$$

Absence of an unstable band does not preclude a finite $\hbar_B \lambda_g u^2$ renormalisation: it shifts both $s_{1,g}$ and C_g by λ_g. If the physical sound speed is defined to remain one, the corresponding finite coefficient is $\lambda_g = 9/(4\pi)$ and the soliton slope relative to that sound line is $+1/(4\pi)$. With the bare Hamiltonian and the normal-ordering condition fixed, no such extra subtraction is

required. **(d)** Toda in the present real scaling regulates the bad branch, so it does not force the same correction on the good equation. Neither a common finite-amplitude quantum completion nor the remaining good projected determinant domain follows from this calculation. "Scheme independent" is justified only after fixing the acoustic renormalisation and boundary conditions, not from linear well-posedness alone.

5 Verdict, reproducibility and failures recorded

Audited claim	Assessment	Check
Failed $3\,10^{-6}$ assertion	**(c)** Incorrect numerical accuracy claim; limiting identity confirmed.	edge
Six dictionary factors	**(c)** Confirmed at leading classical order; do not exclude finite quantum matching.	dictionary
$C_T = 3/(4\pi) - 1/(4\pi)$	**(c)** Confirmed for the nested edge limit; uniform fixed-P interchange unproved (d).	all three
$C_{\mathrm{ac}} = 2/\pi$	**(c)** Confirmed as conditional continued trace; physical determinant identification (d).	continuum
Light-root cancellation / all short-wave origin	**(c)** Log weight vanishes; literal spectral-localization inference is unsupported (d).	continuum
Continuum sound unrenormalised	**(c)** Refuted: good $-9/(4\pi)$, continued bad $+9/(4\pi)$.	continuum
Scheme A equals Toda after a tadpole change	**(c)** Refuted for the stated trace; nonzero even threshold term remains.	continuum
Matching prescription	**(c)** $\Delta\mathcal{H} = -3\hbar_B u^2/(2\pi)$, fixed by lattice vacuum elasticity, gives $C = 1/(2\pi)$.	continuum
Genuinely physical factor four established	**(c)** Rejected as an inference: the acoustic baselines were not matched. Full quantum equivalence (d).	all three

The check column abbreviates script names: add the prefix `onebody_scheme_` and the suffix `.py`. Each has a companion JSON with values, precision, and relevant limitations. Every formula newly claimed above as (c) is checked by symbolic algebra, independent quadrature, or the numerical asymptotics in these scripts; the stated physical and uniform-limit qualifications are not promoted to numerically proved theorems.

Conclusion. **(c)** Choose **(A) for the factor-of-four coefficient**: the lattice-compatible prescription (29) is identified and reproduces C_T exactly, conditional on the inherited continued trace. It is a finite vacuum-elastic matching prescription, not an arbitrary rescaling of $\hbar$ or a mass tadpole. **(d)** If (A) is instead meant to assert equality of fully defined continuum and lattice quantum theories, that stronger question is undetermined: the physical bad contour/domain, boundary-compatible projected determinant, and a uniform quantum matching construction are still missing. No claim of (B) follows from the audited calculations.

Failures recorded.

1. The unchanged original audit failed at line 383. The measured maximum residual is $2.2546734882\,10^{-4}$; three rows ($y = 16, 32, 64$) fail its $3\,10^{-6}$ bound. Its statement of agreement below $2.8\,10^{-6}$ is false for the files inspected. Nothing in that file was changed.

2. The fresh composite sea with 16-point panels at $h = .02$ failed the endpoint-product tolerance $2\,10^{-11}$ with residual $3.3117711365\,10^{-8}$. A 24-point retry also failed, at $4.4067346994\,10^{-11}$. The final calculation uses 32-point panels and a 40-point refinement; the original tolerance is retained.

3. The finer cubic fit still fails $3\,10^{-6}$ at $y = 64$. Quartic and quintic fits are explicitly reported separately; a successful higher-order fit is not relabelled as success of the cubic or old fit.

4. The audit's $b = \pi/a$, intermediate velocity subtraction, energy e_0/y^2 tail, and resonance kinematics fail the independent algebra as detailed above. The zero continuum sound shift fails the vacuum curvature calculation. The broad short-wave and physical-discrepancy conclusions are therefore qualified rather than inherited.

5. The specified stable-band trace has a divergent soft ratio and is not repaired by the local elastic matching term. We retain that failed matching candidate instead of removing its even threshold contribution.

6. The initial PDF build reported two overfull lines. The transform display and reproduction paragraph were reflowed; the final two-pass build has no overfull lines or unresolved references. An auxiliary log-check assertion initially mistook the package description "Rerun checks for auxiliary files" for a rerun warning. Inspection identified that false positive; the corrected check searches warning records instead.

Reproduction and scope of execution. Run the three new Python scripts from their directory, then run `pdflatex -interaction=nonstopmode -halt-on-error onebody_scheme.tex` twice in `phase2/`. Every new script ends with `ALL CHECKS PASSED`. The original audit's expected failure is recorded separately. No repository-wide historical sweep was run: existing scripts can write non-prefix outputs, forbidden by this task. Existing input data were read only; the edge script records and rechecks their SHA-256 digests.

6 Internal sources read

No external literature is used as an independently verified authority here. The internal documents used are `audit_onebody.tex`, `audit_det_projected.tex`, `det_projected_oneloop.tex`, `onebody_dispersion_test.tex`, `onebody_toda_precision.tex`, `toda_route.tex`, and relevant sections of `oneloop.tex` and `frame_dictionary.tex`. The old audit, one-body and frame analysis scripts, Toda classical/scaling routines, and precision edge/sea implementations were read to locate conventions and tests. Their mathematical inputs are rechecked above without importing their code. Saved finite-sea data are identified explicitly where reused; fresh calculations are distinguished from saved-data reanalysis.